

Lecture 05

Linear Maps

Rank-Nullity Theorem

Isomorphisms

Matrix Representation

MH1201 Linear Algebra 2

Lecture 05 – Linear Maps

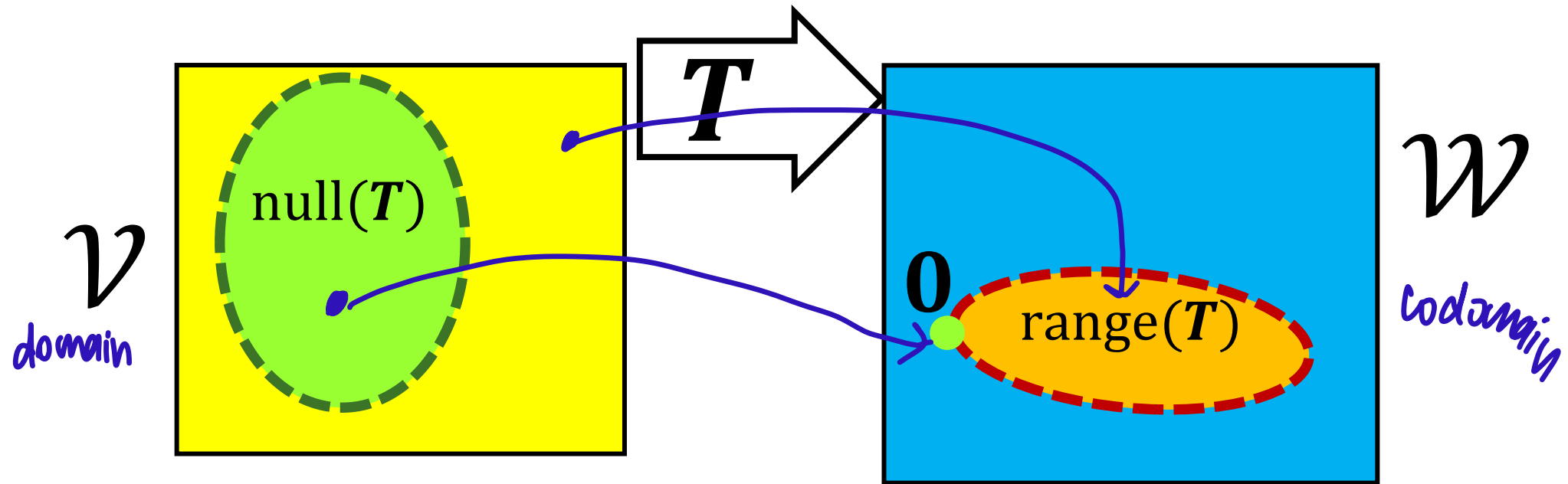
- Applications of Rank-Nullity Theorem
- Rank-Nullity Theorem
- Isomorphisms
- Matrix Representation of Linear Maps
 - Compositions of Linear Maps
 - Finding Range using Matrix Representations
 - Finding Nullspace using Matrix Representations

Applications of Rank-Nullity Theorem

Review. Linear Map

$$\text{null}(\mathbf{T}) = \{\mathbf{v} \in \mathcal{V} : \mathbf{T}(\mathbf{v}) = \mathbf{0}\}$$

$$\text{range}(\mathbf{T}) = \{\mathbf{T}(\mathbf{v}) : \mathbf{v} \in \mathcal{V}\}$$



Rank-Nullity Theorem. Let $\mathcal{V}$ be a finite-dimensional space.

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map. Then

involves only numbers $\left\{ \dim \text{null}(\mathbf{T}) + \dim \text{range}(\mathbf{T}) = \dim(\mathcal{V}) \right.$

Recall: Proposition. (Conditions for Injectivity)

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

→ proved in last lecture

T is injective iff $\text{null}(T) = \{\mathbf{0}\}$ iff $\dim \text{null}(T) = 0$.

*↓
This is because
 $\dim \{\mathbf{0}\} = 0$*

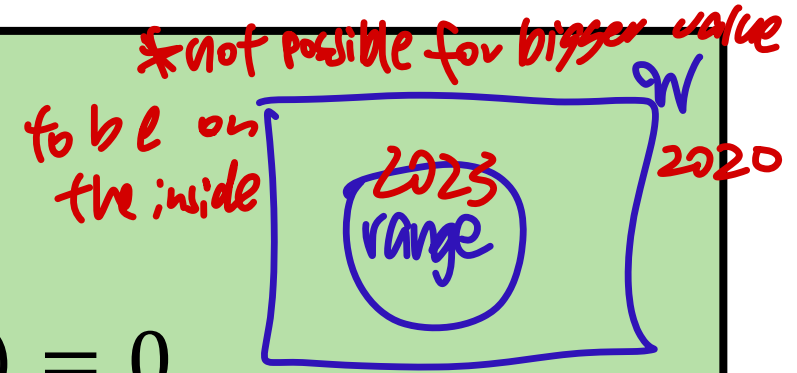
** $\{\mathbf{0}\}$ aka trivial space*

** $\dim \{u\} = 0$ iff $u = \{\mathbf{0}\}$*

Recall: Proposition. (Conditions for Injectivity)

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

T is injective iff $\text{null}(T) = \{\mathbf{0}\}$ iff $\dim \text{null}(T) = 0$.



Proposition.

Let $\mathcal{V}$ be a finite-dimensional space.

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map
with $\dim(\mathcal{V}) > \dim(\mathcal{W})$.
e.g. 2023 > 2020

Can T be injective?

No

Suppose $\bar{T}$ is injective $\Rightarrow \dim \text{null}(\bar{T}) = 0$

Rank Nullity Theorem.

$$\dim \text{null}(\bar{T}) + \dim \text{range}(\bar{T}) = \dim(\mathcal{V})$$

0 ? 2023

$\therefore \text{range}(\bar{T})$ has dimension 2023.

But $\text{range}(\bar{T}) \subseteq \mathcal{W} \nsubseteq \dim(\mathcal{W}) = 2020$

Contradiction

Recall: Proposition. (Conditions for Injectivity)

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

T is injective iff $\text{null}(T) = \{\mathbf{0}\}$ iff $\dim \text{null}(T) = 0$.

Proposition.

Let $\mathcal{V}$ be a finite-dimensional space.

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map
with $\dim(\mathcal{V}) > \dim(\mathcal{W})$.

Then T is not injective.

Let us determine $\dim(\text{null}(T))$

$$\dim \text{null}(T) = \dim \mathcal{V} - \dim \text{range}(T)$$

$$\geq \dim \mathcal{V} - \dim(\mathcal{W})$$

$$> 0$$

$\therefore T$ is not injective.

rank nullity
theorem

$\text{range}(T) \subseteq \mathcal{W}$
So, $\dim(\text{range}(T)) \leq \dim(\mathcal{W})$

Recall: Definition/Proposition.

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

T is surjective iff $\text{range}(T) = \mathcal{W}$ iff $\dim \text{range}(T) = \dim \mathcal{W}$.

↓
definition

↓
this is because...

Prop. If $\mathcal{U} \subseteq \mathcal{W}$ and $\dim(\mathcal{U}) = \dim(\mathcal{W})$, then $\mathcal{U} = \mathcal{W}$.

Recall: Definition/Proposition.

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

T is surjective iff $\text{range}(T) = \mathcal{W}$ iff $\dim \text{range}(T) = \dim \mathcal{W}$.

Proposition.

Let $\mathcal{V}$ be a finite-dimensional space.

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map
with $\dim(\mathcal{V}) < \dim(\mathcal{W})$.

Then T is not surjective.

$$\dim \text{range}(T) = \dim \mathcal{V} - \dim \text{null}(T)$$

$$\leq \dim \mathcal{V}$$

$$< \dim \mathcal{W}$$

$\therefore T$ is not surjective

Rank
Nullity
Theorem

Proposition. Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional spaces.

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map such that

T is both injective and surjective.

What can we say about $\dim(\mathcal{V})$ and $\dim(\mathcal{W})$?

Rank-Nullity Theorem. (Linear Algebra 1 version)

Let A be an $m \times n$ matrix. Then

$$\dim \text{null space}(A) + \dim \text{column space}(A) = n .$$

Prove RN Theorem (LA1) using RN Theorem (LA2 version).

Rank-Nullity Theorem. (Linear Algebra 2 version)

Let $\mathcal{V}$ be a finite-dimensional space.

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map. Then

$$\dim \text{null}(T) + \dim \text{range}(T) = \dim(\mathcal{V})$$

Recall: Proposition. (Matrix Multiplication is a Linear Map)

Let A be an $m \times n$ matrix.

If $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is defined by $T(v) = Av$, then T is a linear map. $\rightarrow n=2$

Let us call it a **matrix-multiplication map**.

What is $\text{null}(T)$ and $\text{range}(T)$?

Set $A = \begin{bmatrix} -1 & 2 \\ 0 & 0 \\ 1 & -2 \end{bmatrix}$ $m=3$

$$\begin{aligned} \text{null}(T) &= \{x \in \mathbb{R}^n : \hat{A} \tilde{x} = \tilde{0}\} \\ &= \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in \mathbb{R}^2 : \begin{bmatrix} -1 & 2 \\ 0 & 0 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \tilde{0} \right\} \end{aligned}$$

$$= \text{nullspace of } \begin{bmatrix} -1 & 2 \\ 0 & 0 \\ 1 & -2 \end{bmatrix} \leftarrow \text{Definition of null space in CA1}$$

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$$

$$\hat{T} \begin{bmatrix} x \\ y \end{bmatrix} = \hat{A} \begin{bmatrix} x \\ y \end{bmatrix}$$

Recall: Proposition. (Matrix Multiplication is a Linear Map)

Let A be an $m \times n$ matrix.

If $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is defined by $T(v) = Av$, then T is a linear map.

Let us call it a **matrix-multiplication map**.

$$\text{Set } A = \begin{bmatrix} -1 & 2 \\ 0 & 0 \\ 1 & -2 \end{bmatrix}$$

What is $\text{null}(T)$ and $\text{range}(T)$?

$$\begin{aligned} \text{range}(T) &= \text{span} \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix} \right\} \\ &= \text{span} \left\{ \begin{bmatrix} -1 & 2 \\ 0 & 0 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 & 2 \\ 0 & 0 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} \\ &= \text{span} \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix} \right\} \\ &= \text{column space of } A \leftarrow \text{defined in LA1} \end{aligned}$$

$$\text{Basis for } \mathbb{R}^2 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

Proposition. (Matrix Multiplication is a Linear Map)

Let A be an $m \times n$ matrix.

Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be the associated matrix-multiplication map, defined by $T(v) = Av$. Then

$$\text{null}(T) = \text{null space of } A \text{ and } \text{range}(T) = \text{column space of } A.$$

Rank-Nullity Theorem. (Linear Algebra 1 version)

Let A be an $m \times n$ matrix. Then

$$\dim \text{null space}(A) + \dim \text{column space}(A) = n .$$

Proof: Apply Rank Nullity Theorem (Linear Map Version)

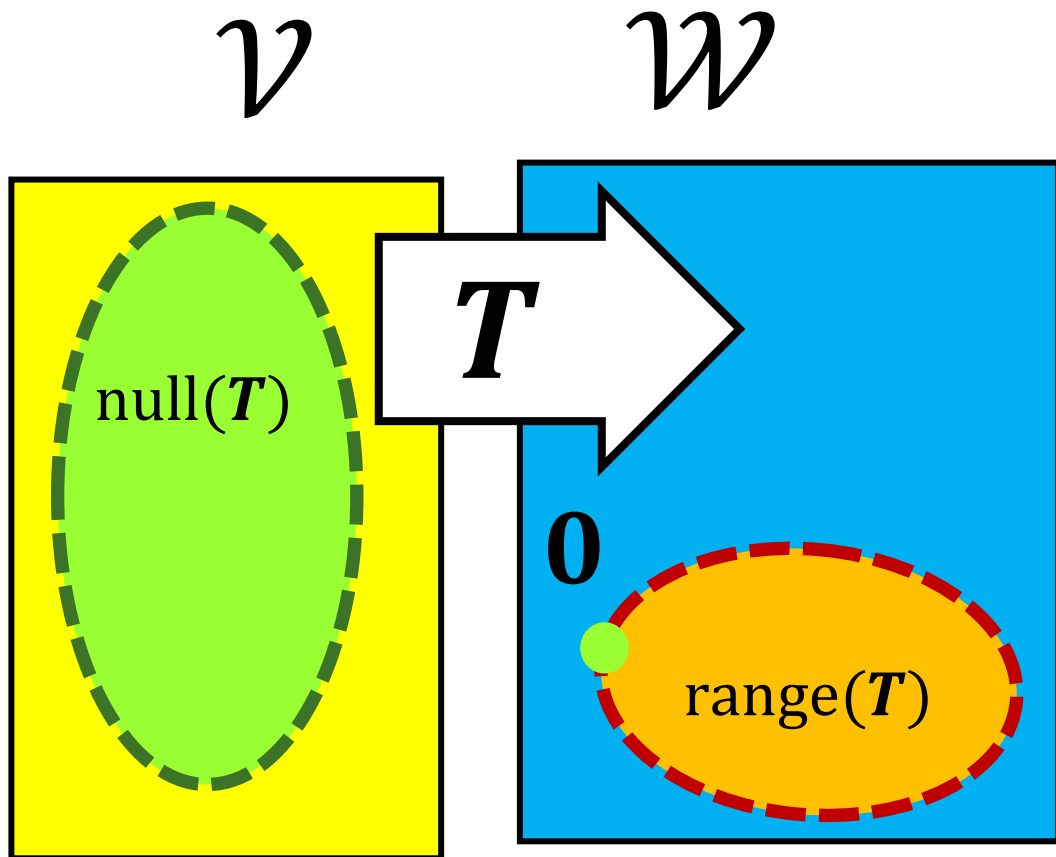
for the matrix multiplication map $T(v) = Av$

Proof of Rank-Nullity Theorem

Rank-Nullity Theorem. Let $\mathcal{V}$ be a finite-dimensional space.

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map. Then

$$\dim \overset{s}{\text{null}}(T) + \dim \overset{t}{\text{range}}(T) = \dim(\overset{\rightarrow \text{domain}}{\mathcal{V}})$$



Proof Plan

1. Construct a **basis** for **null(T)**
 $\rightarrow B$ with $|B| = s$
2. Construct a **basis** for **range(T)**
 $\rightarrow C$ with $|C| = t$
 $\hookrightarrow$ using the basis for null(T)
i.e. using B to construct C
3. Hope that the “numbers add up”
i.e. $s + t = \dim(\mathcal{V})$

Example. Consider the linear map $T: \mathbb{R}^3 \rightarrow \mathbb{R}$ with

$$T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = x + 2y + 3z. \quad \begin{bmatrix} -2s - 3t \\ s \\ t \end{bmatrix}$$

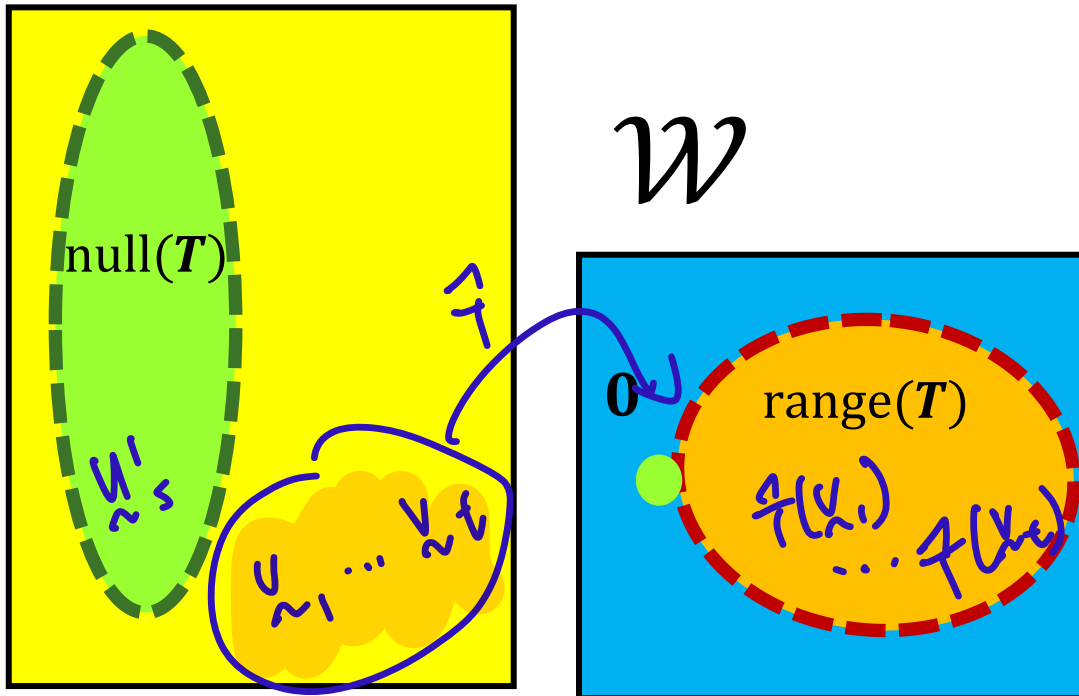
$$\text{Then } \text{null}(T) = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x = -2s - 3t, y = s, z = t, \text{ where } s, t \in \mathbb{R} \right\} = \left\{ s \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} : s, t \in \mathbb{R} \right\}$$

And $\text{range}(T) = \mathbb{R}$. *Question: Does this procedure always work?*

1. So, a basis for $\text{null}(T) = \left[\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right] \triangleq \mathcal{B}$ Now, $\mathcal{B}$ is NOT a basis for $\mathbb{R}^3$.
2. Extend $\mathcal{B}$ to a basis for $\mathbb{R}^3$ $\left[\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right]$
3. Let $\mathcal{C} = \left[\hat{T} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right] = 1$ ← this is a basis for $\mathbb{R}$ (the range)

Construction of Bases

$\mathcal{V}$



Rank-Nullity Theorem. Let $\mathcal{V}$ be a finite-dimensional space.

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map. Then

$$\dim \text{null}(T) + \dim \text{range}(T) = \dim(\mathcal{V})$$

1. Construct a basis for $\text{null}(T)$. Let this basis be

$$\mathcal{B} = [u_1, \dots, u_s].$$

2. Extend this basis to obtain a basis for $\mathcal{V}$

$$[u_1, \dots, u_s, v_1, \dots, v_t].$$

← this is possible because of Basis Extension Theorem

3. Consider

$$\mathcal{C} = [T(v_1), \dots, T(v_t)]$$

4. Hope that $\mathcal{C}$ is a basis for $\text{range}(T)$.

$$\rightarrow \text{span}(\mathcal{C}) = \text{range}(T)$$

$\rightarrow \mathcal{C}$ is linearly independent.

Proof Continued

Claim. $\{T(v_1), \dots, T(v_t)\}$ is a basis for $\text{range}(T)$.

→ check $\hat{T}(u_1) \dots \hat{T}(u_s)$ span $\text{range}(\hat{T})$

Because $[u_1 \dots u_s, v_1 \dots v_t]$ is a basis for $\mathcal{V}$,
 $\text{range}(\hat{T}) = \text{span}(\hat{T}(u_1) \dots \hat{T}(u_s), \hat{T}(v_1) \dots \hat{T}(v_t))$
 $= \text{span}(\mathbf{0} \dots \mathbf{0}, \hat{T}(v_1) \dots \hat{T}(v_t))$
 $= \text{span}(\hat{T}(v_1) \dots \hat{T}(v_t))$

Null
Space

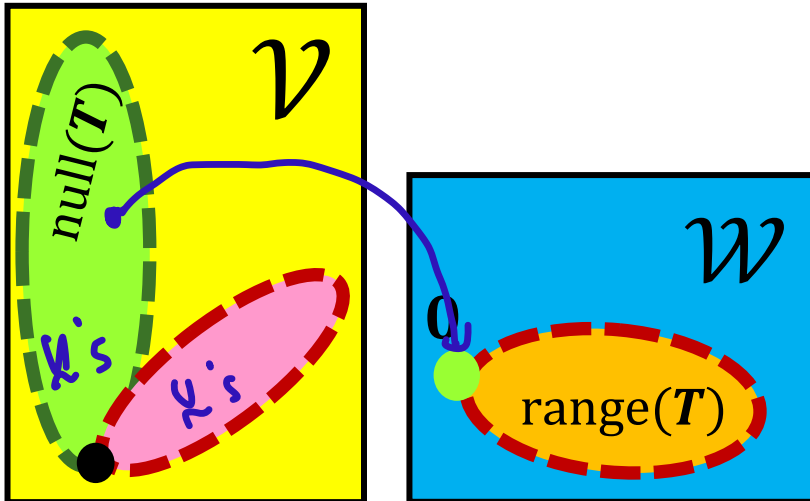
as required.

$\{u_1, \dots, u_s\}$ is a basis for $\text{null}(T)$.

So, $\dim \text{null}(T) = s$.

$\{u_1, \dots, u_s, v_1, \dots, v_t\}$ is a basis for $\mathcal{V}$.

So, $\dim \mathcal{V} = s + t$.

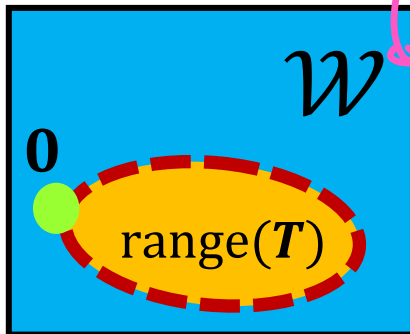
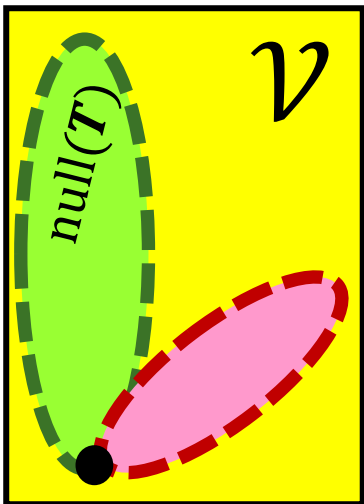


Proof Continued

Claim. $\{T(v_1), \dots, T(v_t)\}$ is a basis for $\text{range}(T)$.

$\Rightarrow \{\hat{T}(v_1), \dots, \hat{T}(v_t)\}$ is linearly independent

- $\{u_1, \dots, u_s\}$ is a basis for $\text{null}(T)$.
- So, $\dim \text{null}(T) = s$.
- $\{u_1, \dots, u_s, v_1, \dots, v_t\}$ is a basis for V .
- So, $\dim V = s + t$.



$$\lambda_1 \hat{T}(v_1) + \dots + \lambda_t \hat{T}(v_t) = 0 \quad \text{--- } (*)$$

$\Rightarrow$ becomes Cyclic Space
 $T(\lambda_1 v_1 + \dots + \lambda_t v_t) = 0$ because $\hat{T}$ is linear

~~Wrong!~~
 $\lambda_1 v_1 + \dots + \lambda_t v_t = 0$ $\rightarrow$ things in nullspace also can go to 0
 $\downarrow$ can write as linear combi of u_i .

because $u_1, \dots, u_s$ are linearly independent and $v_1, \dots, v_t$ are linearly independent
 $\lambda_1 v_1 + \dots + \lambda_t v_t = u_1 u_1 + \dots + u_s u_s$
 $\Rightarrow \lambda_1 v_1 + \dots + \lambda_t v_t - u_1 u_1 - \dots - u_s u_s = 0 \quad \text{--- } (**)$ because u_i belongs to $\text{null}(T)$

$$\Rightarrow \lambda_1 = \lambda_2 = \dots = \lambda_t = u_1 = \dots = u_s = 0$$

$$\Rightarrow \lambda_1 = \dots = \lambda_t = 0$$

Completing the Proof

Rank-Nullity Theorem. Let $\mathcal{V}$ be a finite-dimensional space.

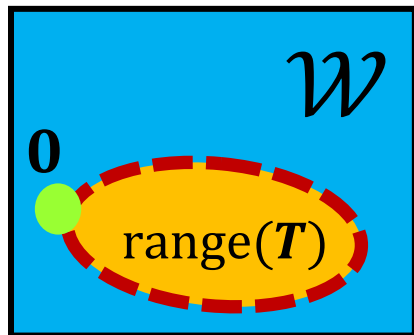
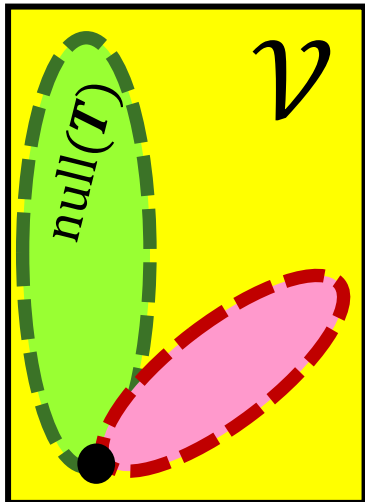
Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map. Then

$$\dim \text{null}(T) + \dim \text{range}(T) = \dim(\mathcal{V})$$

- $\{\mathbf{u}_1, \dots, \mathbf{u}_s\}$ is a basis for $\text{null}(T)$.
- So, $\dim \text{null}(T) = s$.
- $\{\mathbf{u}_1, \dots, \mathbf{u}_s, \mathbf{v}_1, \dots, \mathbf{v}_t\}$ is a basis for $\mathcal{V}$.
- So, $\dim \mathcal{V} = s + t$.

$[\hat{T}(\mathbf{v}_1) \dots \hat{T}(\mathbf{v}_t)]$ is a basis for $\text{range}(\hat{T})$
 $\Rightarrow \dim \text{range}(\hat{T}) = t$

$$\begin{aligned} \dim \mathcal{V} &= \dim \text{null}(\hat{T}) + \dim \text{range}(\hat{T}) \\ &= s + t \\ &= \dim \mathcal{V} = \text{RHS} \end{aligned}$$



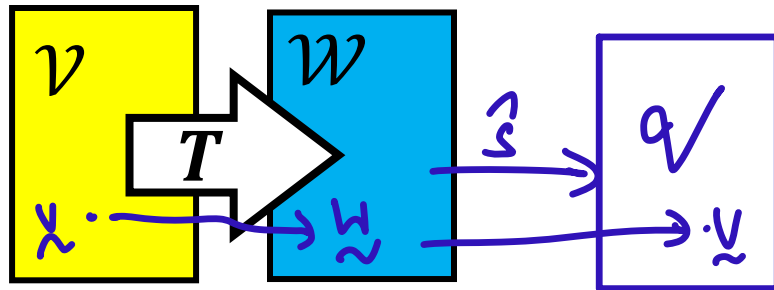
Isomorphisms aka Invertible Linear Maps

Definition. Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

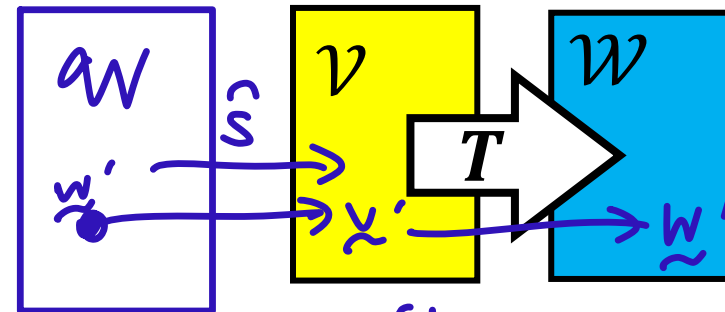
T is invertible if there exists a map $S: \mathcal{W} \rightarrow \mathcal{V}$ such that

$$ST = I_{\mathcal{V}} \text{ and } TS = I_{\mathcal{W}}. \quad \hat{I} \text{ is the identity map.}$$

If S satisfies the above conditions, we say that S is an inverse of T .



$$ST = I_V$$



$$TS = I_W$$

Example. Consider the linear map $T: \mathcal{P}_2(\mathbb{R}) \rightarrow \mathbb{R}^3$ with

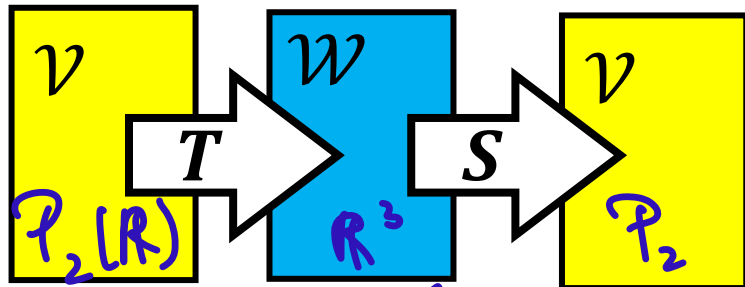
$$T(a + bx + cx^2) = \begin{bmatrix} a \\ b \\ c \end{bmatrix}.$$

Then T is invertible because...

...Set $\hat{S}: \mathbb{R}^3 \rightarrow \mathcal{P}_2(\mathbb{R})$

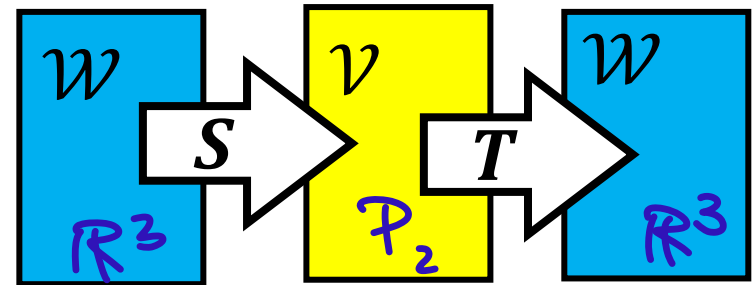
$$\hat{S} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = a + bx + cx^2$$

And so, $\hat{S}$ is an inverse of $\hat{T}$



check that: $\hat{S}\hat{T} = I_V$
 $\hat{S}\hat{T}(a+bx+cx^2) = \hat{S} \begin{bmatrix} a \\ b \\ c \end{bmatrix}$
 $= a + bx + cx^2$

AND



check that $\hat{T}\hat{S} = I_W$
 $\hat{T}\hat{S} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \hat{T}(a+bx+cx^2)$
 $= \begin{bmatrix} a \\ b \\ c \end{bmatrix}$

Definition. Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

T is invertible if there exists a map $S: \mathcal{W} \rightarrow \mathcal{V}$ such that

$$ST = I_{\mathcal{V}} \text{ and } TS = I_{\mathcal{W}}.$$

If S satisfies the above conditions, we say that S is an inverse of T .

Proposition. Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

If T is invertible, then the inverse map is unique.

So we write

$$S = T^{-1}$$

Proof omitted

(this is really an exercise from FOM)

Example. Consider the linear map $T: \mathcal{P}_2(\mathbb{R}) \rightarrow \mathbb{R}^3$ with

$$T(a + bx + cx^2) = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

and the map $S: \mathbb{R}^3 \rightarrow \mathcal{P}_2(\mathbb{R})$ with

$$S\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = a + bx + cx^2.$$

Then S is ^{the} an inverse of T .

and we can write:

$$T^{-1}\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = a + bx + cx^2$$

Unique here means that if $\hat{S}'$ is also an inverse of T , then
 $\hat{S}'\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = a + bx + cx^2$

Definition. Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

T is invertible if there exists a map $S: \mathcal{W} \rightarrow \mathcal{V}$ such that

$$ST = I_{\mathcal{V}} \text{ and } TS = I_{\mathcal{W}}.$$

If S satisfies the above conditions, we say that S is the inverse of T .

Proposition. Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

If T is invertible, then the inverse map T^{-1} is linear.

Proof left as an exercise.

Check that T^{-1} is additive & homogeneous

Example. Consider the linear map $T: \mathcal{P}_2(\mathbb{R}) \rightarrow \mathbb{R}^3$ with

$$T(a + bx + cx^2) = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

and the map $S: \mathbb{R}^3 \rightarrow \mathcal{P}_2(\mathbb{R})$ with

$$S\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = a + bx + cx^2.$$

Then S is the inverse of T .

Exercise : Show that $\hat{J}$ is linear

Definition. Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

T is invertible if there exists a map $S: \mathcal{W} \rightarrow \mathcal{V}$ such that

$$ST = I_{\mathcal{V}} \text{ and } TS = I_{\mathcal{W}}.$$

If S satisfies the above conditions, we say that S is the inverse of T .

Proposition. Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

T is invertible if and only if T is both injective and surjective.

- Proof omitted
- This is really an exercise from FOM

Example. Consider the linear map $T: \mathcal{P}_2(\mathbb{R}) \rightarrow \mathbb{R}^3$ with

$$T(a + bx + cx^2) = \begin{bmatrix} a \\ b \\ c \end{bmatrix}.$$

Then T is invertible.

zero polynomial

Exercise: Show that $\text{null}(T) = \{z(x)\} \Rightarrow T$ is injective
and $\text{range}(T) = \mathbb{R}^3 \Rightarrow T$ is surjective

Definition. Let $\mathcal{V}$ and $\mathcal{W}$ be two vector spaces. $\mathcal{V}$ and $\mathcal{W}$ are said to be isomorphic if there is an invertible linear map $\phi: \mathcal{V} \rightarrow \mathcal{W}$. ϕ is also called an isomorphism.

Proposition. $\mathcal{V}$ and $\mathcal{W}$ are isomorphic if and only if $\dim(\mathcal{V}) = \dim(\mathcal{W})$.

Proof ($\mathcal{V}$ and $\mathcal{W}$ are isomorphic $\Rightarrow \dim \mathcal{V} = \dim \mathcal{W}$)

• There exists an invertible $\phi: \mathcal{V} \rightarrow \mathcal{W}$. So ϕ is both injective & surjective.

Rank Nullity Theorem: $\dim \text{Null } \phi + \dim \text{range } \phi = \dim \mathcal{V}$.

$\Downarrow \Rightarrow \because \phi$ is injective

$\Downarrow \dim \mathcal{W} \because \phi$ is surjective

$\therefore \dim \mathcal{W} = \dim \mathcal{V}$

Definition. Let $\mathcal{V}$ and $\mathcal{W}$ be two vector spaces. $\mathcal{V}$ and $\mathcal{W}$ are said to be isomorphic if there is an invertible linear map $\phi: \mathcal{V} \rightarrow \mathcal{W}$. ϕ is also called an isomorphism.

the other direction
Proposition. $\mathcal{V}$ and $\mathcal{W}$ are isomorphic if and only if $\dim(\mathcal{V}) = \dim(\mathcal{W})$.

Proof $\dim \mathcal{V} = \dim \mathcal{W} \Rightarrow \mathcal{V}$ and $\mathcal{W}$ are isomorphic

— *Proof omitted.*

Essentially, we need to construct an invertible linear map $T: \mathcal{V} \rightarrow \mathcal{W}$.

Definition. Let $\mathcal{V}$ and $\mathcal{W}$ be two vector spaces. $\mathcal{V}$ and $\mathcal{W}$ are said to be isomorphic if there is an invertible linear map $\phi: \mathcal{V} \rightarrow \mathcal{W}$. ϕ is also called an isomorphism.

Proposition. $\mathcal{V}$ and $\mathcal{W}$ are isomorphic if and only if $\dim(\mathcal{V}) = \dim(\mathcal{W})$.

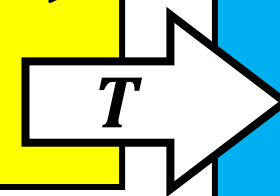
Example. Consider the linear map $T: \mathcal{P}_2(\mathbb{R}) \rightarrow \mathbb{R}^3$ with

$$T(a + bx + cx^2) = \begin{bmatrix} a \\ b \\ c \end{bmatrix}.$$

Then T is invertible. and T is an isomorphism. By definition, $\mathcal{P}_2$ and $\mathbb{R}^3$ are isomorphic.

Check: $\dim \mathcal{P}_2 = 3$
 $\dim \mathbb{R}_3 = 3$

$\mathcal{V} = \mathcal{P}_2(\mathbb{R})$



$\mathcal{W} = \mathbb{R}^3$

Proposition. Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map where $\mathcal{V}$ is finite-dimensional and $\dim(\mathcal{V}) = \dim(\mathcal{W})$. Then the following are equivalent:

(a) T is invertible;

(b) T is injective;

(c) T is surjective.

To check Invertibility, if $\dim(\mathcal{V}) = \dim(\mathcal{W})$, just need to

check either T is injective or surjective.

$\rightarrow$ just check one thing.

implies

Recall : $(a) \Rightarrow [(b) \wedge (c)]$

$\left. \begin{array}{l} (b) \Rightarrow (c) \\ (c) \Rightarrow (b) \end{array} \right\} \Rightarrow (a)$

It suffices to show that $(b) \Rightarrow (c)$

T injective

$$\begin{aligned} \dim \text{range}(T) &= \dim \mathcal{V} - \dim \text{null } T \\ &= \dim \mathcal{V} \\ &= \dim \mathcal{W} \end{aligned}$$

$\Rightarrow T$ surjective

It suffices to show that $(c) \Rightarrow (b)$

T is surjective

$$\begin{aligned} \dim \text{null } T &= \dim \mathcal{V} - \dim \text{range } T \\ &= \dim \mathcal{V} - \dim \mathcal{W} \\ &= 0 \end{aligned}$$

$\Rightarrow T$ injective

Matrix Representations

Definition (Recall).

Let $B = \{v_1, \dots, v_n\}$ be a basis for V .

Then the *coordinate vector* of v with respect to basis B , denoted by $[v]_B$ is the tuple $(\lambda_1, \dots, \lambda_n)^T$ such that

$$v = \lambda_1 v_1 + \dots + \lambda_n v_n.$$

Definition. Let $T : V \rightarrow W$ be a linear map.

Let $B = \{v_1, \dots, v_n\}$ be a basis for V .

Let $\mathcal{A} = \{w_1, \dots, w_m\}$ be a basis for W .

Then the *matrix representation with respect to bases $\mathcal{A}$ and B* , denoted by $[T]_B^{\mathcal{A}}$, is the matrix such that

$$T(v_j) = T_{1,j} w_1 + \dots + T_{m,j} w_m.$$

Let $V = \mathcal{P}_2(\mathbb{R})$ & $B = [1, x, x^2]$

Then $[a + bx + cx^2]_B = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ ¹ _{x} _{x^2}

Because $a + bx + cx^2 = a + bx + cx^2$

— Coordinate vector is a representation of a vector as a tuple.

— Matrix representation is a representation of a linear map as a matrix

Definition. Let $T : \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

Let $\mathcal{B} = \{v_1, \dots, v_n\}$ be a basis for $\mathcal{V}$.

Let $\mathcal{A} = \{w_1, \dots, w_m\}$ be a basis for $\mathcal{W}$.

Then the *matrix representation with respect to bases $\mathcal{A}$ and $\mathcal{B}$* , denoted by $[T]_{\mathcal{B}}^{\mathcal{A}}$, is the matrix such that

$$T(v_j) = T_{1,j}w_1 + \dots + T_{m,j}w_m.$$

Don't memorise!

A

max matrix

basis for codomain

basis for domain

$$\begin{matrix} \begin{matrix} w_1 \\ \vdots \\ w_m \end{matrix} & \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}^A & \\ \underbrace{\begin{matrix} v_1 & \dots & v_n \end{matrix}}_{\mathcal{B}} & & \end{matrix}$$

Example. Consider the differentiation map

$D_3 : \mathcal{P}_3(\mathbb{R}) \rightarrow \mathcal{P}_2(\mathbb{R})$ with $D_3(f) = f'$.

What is $[D_3]_{\mathcal{B}}^{\mathcal{A}}$?

Here, $\mathcal{B} = [1, x, x^2, x^3]$

and $\mathcal{A} = [1, x, x^2]$.

$$\hat{D}_3(1) = 0 + 0x + 0x^2$$

$$\hat{D}_3(x) = 1 + 0x + 0x^2$$

$$\hat{D}_3(x^2) = 2x$$

$$\hat{D}_3(x^3) = 3x^2$$

A

$$\begin{matrix} \begin{matrix} 1 \\ x \\ x^2 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} & = & \begin{bmatrix} \hat{D}_3 \end{bmatrix}_{\mathcal{B}}^{\mathcal{A}} \\ \begin{matrix} 1 & x & x^2 & x^3 \end{matrix} & \leftarrow \mathcal{B} & & \end{matrix}$$

Definition. Let $T : \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

Let $\mathcal{B} = \{v_1, \dots, v_n\}$ be a basis for $\mathcal{V}$.

Let $\mathcal{A} = \{w_1, \dots, w_m\}$ be a basis for $\mathcal{W}$.

Then the *matrix representation with respect to bases $\mathcal{A}$ and $\mathcal{B}$* , denoted by $[T]_{\mathcal{B}}^{\mathcal{A}}$, is the matrix such that

$$T(v_j) = T_{1,j}w_1 + \dots + T_{m,j}w_m.$$

$$[\hat{S}_2]_{\mathcal{B}}^{\mathcal{A}} = \begin{matrix} & 1 & & \\ \begin{matrix} \hat{S}_2 \end{matrix} & \begin{matrix} x \\ x^2 \\ x^3 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

Example. Consider the shift map

$S_2 : \mathcal{P}_2(\mathbb{R}) \rightarrow \mathcal{P}_3(\mathbb{R})$ with $S_2(f) = xf$.

What is $[S_2]_{\mathcal{B}}^{\mathcal{A}}$?

Here, $\mathcal{B} = \{1, x, x^2\}$

and $\mathcal{A} = \{1, x, x^2, x^3\}$.

$$\hat{S}_2(1) = x$$

$$\hat{S}_2(x) = x^2$$

$$\hat{S}_2(x^2) = x^3$$

Matrix Representation of Linear Maps

$$\mathcal{B} = [1, x, x^2, x^3]$$

$$\mathcal{V} = \mathcal{P}_3(\mathbb{R})$$

$$a + bx + cx^2 + dx^3$$

D_3

$$\mathcal{A} = [1, x, x^2]$$

$$\mathcal{W} = \mathcal{P}_2(\mathbb{R})$$

$$b + 2cx + 3dx^2$$

$$\hat{\phi}(v) = \begin{bmatrix} v \\ v \\ v \\ v \end{bmatrix}_{\mathcal{B}}, \quad \hat{\pi}(v) = \begin{bmatrix} v \\ v \\ v \end{bmatrix}_{\mathcal{A}}$$

We see that $\hat{\pi}(\hat{D}_3(f))$ composition of functions

$\downarrow \hat{\phi}$

$$\mathcal{X} = \mathbb{R}^4$$

$$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$$

$$\begin{bmatrix} \hat{D}_3 \end{bmatrix}_{\mathcal{B}}^{\mathcal{A}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

$\downarrow \hat{\pi}$

$$\mathcal{Y} = \mathbb{R}^3$$

$$\begin{bmatrix} b \\ 2c \\ 3d \end{bmatrix}$$

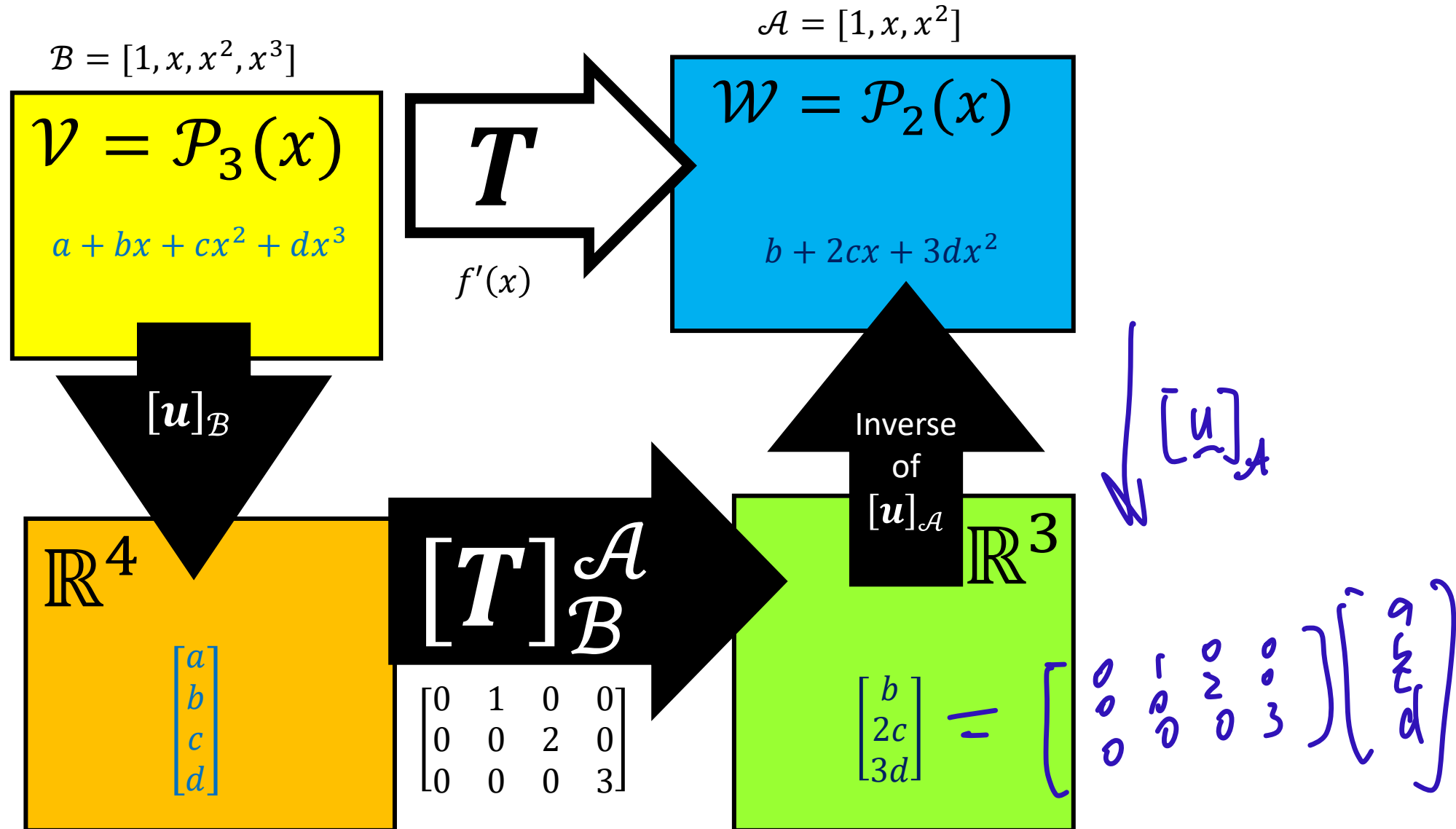
equals

$$= \begin{bmatrix} \hat{D}_3 \end{bmatrix}_{\mathcal{B}}^{\mathcal{A}} \hat{\phi}(f)$$

matrix multiplication

Matrix Representation of Linear Maps

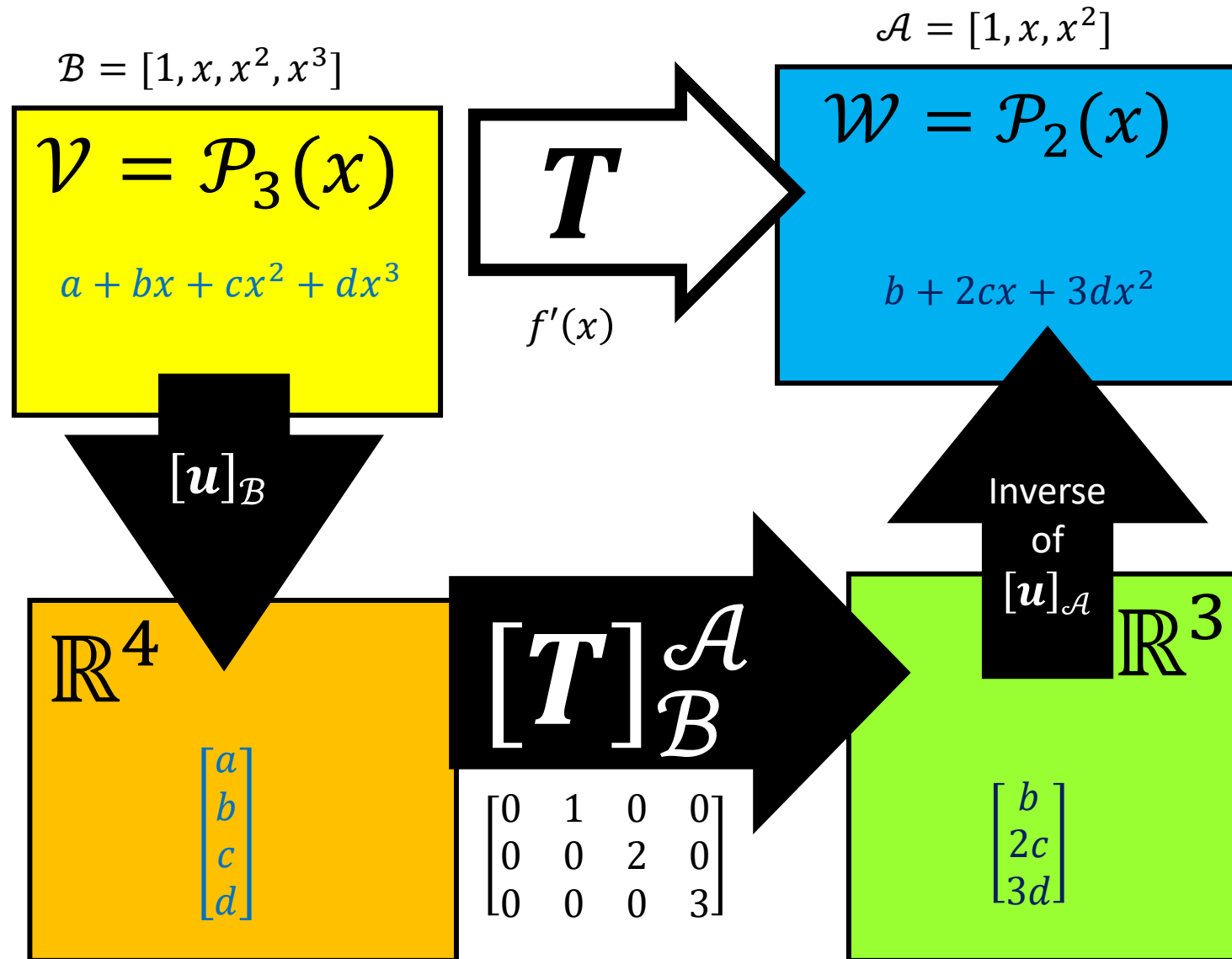
Complete Picture



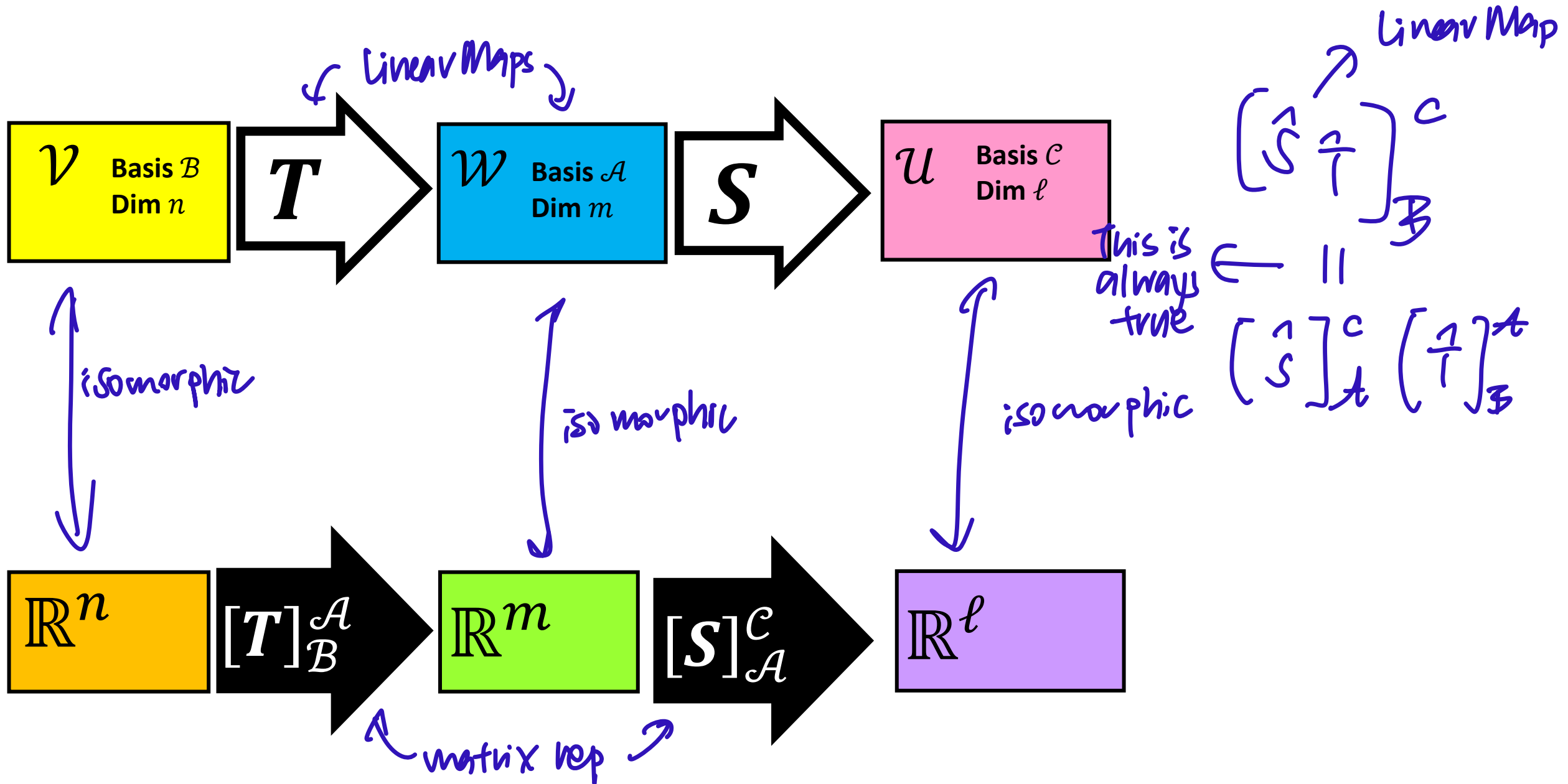
Composition of Linear Maps and Their Matrix Representations

Matrix Representation of Linear Maps

Complete Picture



Composition of Linear Maps



Proposition. composition = matrix multiplication

Let $T : \mathcal{V} \rightarrow \mathcal{W}$ and $S : \mathcal{W} \rightarrow \mathcal{U}$ be linear maps.

Let $\mathcal{B}$ be a basis for $\mathcal{V}$, $\mathcal{A}$ be a basis for $\mathcal{W}$, $\mathcal{C}$ be a basis for $\mathcal{U}$.

Then

$$[ST]_{\mathcal{B}}^{\mathcal{C}} = [S]_{\mathcal{A}}^{\mathcal{C}} [T]_{\mathcal{B}}^{\mathcal{A}}.$$

$\nearrow$ composition of functions $\nearrow$ matrix multiplication

Proof Omitted

$$\begin{bmatrix} \hat{D}_2 & \hat{D}_3 \end{bmatrix}_{\mathcal{B}}^{\mathcal{C}} = \frac{1}{x} \begin{bmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

$1 \quad x \quad x^2 \quad x^3$

Example. Consider the differentiation maps

$D_3 : \mathcal{P}_3(\mathbb{R}) \rightarrow \mathcal{P}_2(\mathbb{R})$ with $D_3(f) = f'$,

$D_2 : \mathcal{P}_2(\mathbb{R}) \rightarrow \mathcal{P}_1(\mathbb{R})$ with $D_2(f) = f'$.

What is $[D_2 D_3]_{\mathcal{B}}^{\mathcal{C}}$?

Here, $\mathcal{B} = \{1, x, x^2, x^3\}$, $\mathcal{A} = \{1, x, x^2\}$

and $\mathcal{C} = \{1, x\}$.

$$\begin{aligned} \hat{D}_2 \hat{D}_3(1) &= 0 \\ \hat{D}_2 \hat{D}_3(x) &= 0 \\ \hat{D}_2 \hat{D}_3(x^2) &= 2 \\ \hat{D}_2 \hat{D}_3(x^3) &= 6x \end{aligned}$$

$$[\hat{D}_2]_{\mathcal{A}}^{\mathcal{C}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$[\hat{D}_3]_{\mathcal{B}}^{\mathcal{A}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

$$[\hat{D}_2][\hat{D}_3] = \begin{bmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

verified that $[\hat{D}_2 \hat{D}_3] = [D_2][D_3]$

Proposition.

Let $T : \mathcal{V} \rightarrow \mathcal{W}$ and $S : \mathcal{W} \rightarrow \mathcal{U}$ be linear maps.

Let $\mathcal{B}$ be a basis for $\mathcal{V}$, $\mathcal{A}$ be a basis for $\mathcal{W}$, $\mathcal{C}$ be a basis for $\mathcal{U}$.

Then

$$[ST]_{\mathcal{B}}^{\mathcal{C}} = [S]_{\mathcal{A}}^{\mathcal{C}} [T]_{\mathcal{B}}^{\mathcal{A}}.$$

Example. Consider the shift maps

$S_1 : \mathcal{P}_1(\mathbb{R}) \rightarrow \mathcal{P}_2(\mathbb{R})$ with $S_1(f) = xf$,

$S_2 : \mathcal{P}_2(\mathbb{R}) \rightarrow \mathcal{P}_3(\mathbb{R})$ with $S_2(f) = xf$.

Check that

$$[S_2 S_1]_{\mathcal{B}}^{\mathcal{C}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

$$\begin{aligned} \mathcal{B} &= [1, x] \\ \mathcal{A} &= [1, x, x^2] \\ \mathcal{C} &= [1, x, x^2, x^3] \end{aligned}$$

DIY

Direct: $\hat{S}_2 \hat{S}_1(f) = x^2 f$

Multiplication: find $[\hat{S}_2]_{\mathcal{A}}^{\mathcal{C}}$

$$\& \begin{bmatrix} 1 \\ \hat{S}_1 \end{bmatrix}_{\mathcal{B}}$$

& multiply the matrices

a basis for

Finding Range

/

Using Matrix Representation

Example. Consider the linear map $T: \mathcal{P}_4(\mathbb{R}) \rightarrow \mathbb{R}^3$ with

$$T(f) = (f(0), f(1), f(2)).$$

Consider the basis $\mathcal{B} = [1, x, x^2, x^3, x^2(x-1)(x-2)]$ for $\mathcal{P}_4(\mathbb{R})$ and $\mathcal{A}$ the standard basis for $\mathbb{R}^3$.

What is $[T]_{\mathcal{B}}^{\mathcal{A}}$?

5 elements

constant function

$$[T]_{\mathcal{B}}^{\mathcal{A}} = \begin{matrix} \begin{matrix} (1,0,0) \\ (0,1,0) \\ (0,0,1) \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 4 & 8 & 0 \\ 1 & x & x^2 & x^3 & x^2(x-1)(x-2) \end{bmatrix} \end{matrix}$$

$$T(1) = (1, 1, 1)$$

$$T(x) = (0, 1, 2)$$

$$T(x^2) = (0, 1, 4)$$

Example. Consider the linear map $T: \mathcal{P}_4(\mathbb{R}) \rightarrow \mathbb{R}^3$ with

$$T(f) = (f(0), f(1), f(2)).$$

Consider the basis $\mathcal{B} = [1, x, x^2, x^3, x^2(x-1)(x-2)]$ for $\mathcal{P}_4(\mathbb{R})$ and $\mathcal{A}$ the standard basis for $\mathbb{R}^3$.

What is a basis for $\text{range}(T)$?

→ without matrix representation

$$\text{range}(T) = \text{span}[T(1), T(x), \dots, T(x^2(x-1)(x-2))]$$

$$= \text{span}[(1, 1, 1), (0, 1, 2), (0, 1, 4), (0, 1, 8), (0, 0, 0)]$$

→ This set is
not linearly
independent

— A basis for $\text{range}(T) = [(1, 1, 1), (0, 1, 2), \dots]$

→ Wrong!

A basis for $\text{range}(T) = [(1, 1, 1), (0, 1, 2), (0, 1, 4)]$ because $[(1, 1, 1), (0, 1, 2), (0, 1, 4)]$
are linearly independent

Example. Consider the linear map $T: \mathcal{P}_4(\mathbb{R}) \rightarrow \mathbb{R}^3$ with

$$T(f) = (f(0), f(1), f(2)).$$

Consider the basis $\mathcal{B} = [1, x, x^2, x^3, x^2(x-1)(x-2)]$ for $\mathcal{P}_4(\mathbb{R})$ and $\mathcal{A}$ the standard basis for $\mathbb{R}^3$.

What is a basis for $\text{range}(T)$?

→ Smith matrix representation

$$[T]_{\mathcal{B}}^{\mathcal{A}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 4 & 8 & 0 \end{bmatrix} \xrightarrow{\text{rref}} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -2 & 0 \\ 0 & 0 & 1 & 3 & 0 \end{bmatrix}$$

Leading entries at col. 1, 2, 3

A basis for range (T)

$$= \left[(1, 1, 1), (0, 1, 2), (0, 1, 4) \right]$$

a basis for

Finding _{*n*} Null Space Using Matrix Representation

Example. Consider the linear map $\mathbf{T}: \mathcal{P}_4(\mathbb{R}) \rightarrow \mathbb{R}^3$ with

$$\mathbf{T}(f) = (f(0), f(1), f(2)).$$

Consider the basis $\mathcal{B} = [1, x, x^2, x^3, x^2(x-1)(x-2)]$ for $\mathcal{P}_4(\mathbb{R})$ and $\mathcal{A}$ the standard basis for $\mathbb{R}^3$.

What is $[\mathbf{T}]_{\mathcal{B}}^{\mathcal{A}}$?

$$[\mathbf{T}]_{\mathcal{B}}^{\mathcal{A}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 4 & 8 & 0 \end{bmatrix}$$

Example. Consider the linear map $T: \mathcal{P}_4(\mathbb{R}) \rightarrow \mathbb{R}^3$ with

$$T(f) = (f(0), f(1), f(2)).$$

Consider the basis $\mathcal{B} = [1, x, x^2, x^3, x^2(x-1)(x-2)]$ for $\mathcal{P}_4(\mathbb{R})$ and $\mathcal{A}$ the standard basis for $\mathbb{R}^3$.

What is a basis for $\text{null}(T)$?

$$\text{null}(T) = \{ f \in \mathcal{P}_4 : (f(0), f(1), f(2)) = (0, 0, 0) \}$$

$$= \{ f \in \mathcal{P}_4 : f(0) = f(1) = f(2) = 0 \}$$

it's like guessing...?

Claim: $\mathcal{C} = [x(x-1)(x-2), x^2(x-1)(x-2)]$ is a basis for $\text{null}(T)$

→ check $\mathcal{C}$ is linearly independent

or → check $\text{span}(\mathcal{C}) = \text{null}(T)$

Example. Consider the linear map $T: \mathcal{P}_4(\mathbb{R}) \rightarrow \mathbb{R}^3$ with

$$T(f) = (f(0), f(1), f(2)).$$

null space $\subseteq$ domain
polynomials

Consider the basis $\mathcal{B} = [1, x, x^2, x^3, x^2(x-1)(x-2)]$ for $\mathcal{P}_4(\mathbb{R})$ and $\mathcal{A}$ the standard basis for $\mathbb{R}^3$.

What is a basis for $\text{null}(T)$?

$$[T]_{\mathcal{B}}^{\mathcal{A}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 4 & 8 & 0 \end{bmatrix} \rightarrow \text{rref} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -2 & 0 \\ 0 & 0 & 1 & 3 & 0 \end{bmatrix}$$

$x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5$

null space of $[T]_{\mathcal{B}}^{\mathcal{A}} = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 2s \\ -3s \\ s \\ t \end{bmatrix} : s, t \in \mathbb{R} \right\}$

$= \left\{ s \begin{bmatrix} 0 \\ 2 \\ -3 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} : s, t \in \mathbb{R} \right\}$

$\left[\begin{bmatrix} 0 \\ 2 \\ -3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right]$
is a basis for

$\text{null}(T)$ ~~Wrong!~~
is supposed to get polynomials

Example. Consider the linear map $T: \mathcal{P}_4(\mathbb{R}) \rightarrow \mathbb{R}^3$ with

$$T(f) = (f(0), f(1), f(2)).$$

Consider the basis $\mathcal{B} = [1, x, x^2, x^3, x^2(x-1)(x-2)]$ for $\mathcal{P}_4(\mathbb{R})$ and $\mathcal{A}$ the standard basis for $\mathbb{R}^3$.

What is a basis for $\text{null}(T)$?

basis for nullspace $\begin{bmatrix} 1 \\ 1 \end{bmatrix}_{\mathcal{B}}^{\mathcal{A}} = \left\{ \begin{bmatrix} 0 \\ 2 \\ -3 \\ 1 \\ 0 \end{bmatrix} \begin{matrix} 1 \\ x \\ x^2 \\ x^3 \\ x^2(x-1)(x-2) \end{matrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

$$\begin{aligned} \text{basis for null}(T) &= [2x - 3x^2 + x^3, x^2(x-1)(x-2)] \\ &= [x(x-1)(x-2), x^2(x-1)(x-2)] \end{aligned}$$